SMARTATTEND

Data Flow Diagram

High-level information flow for SmartAttend

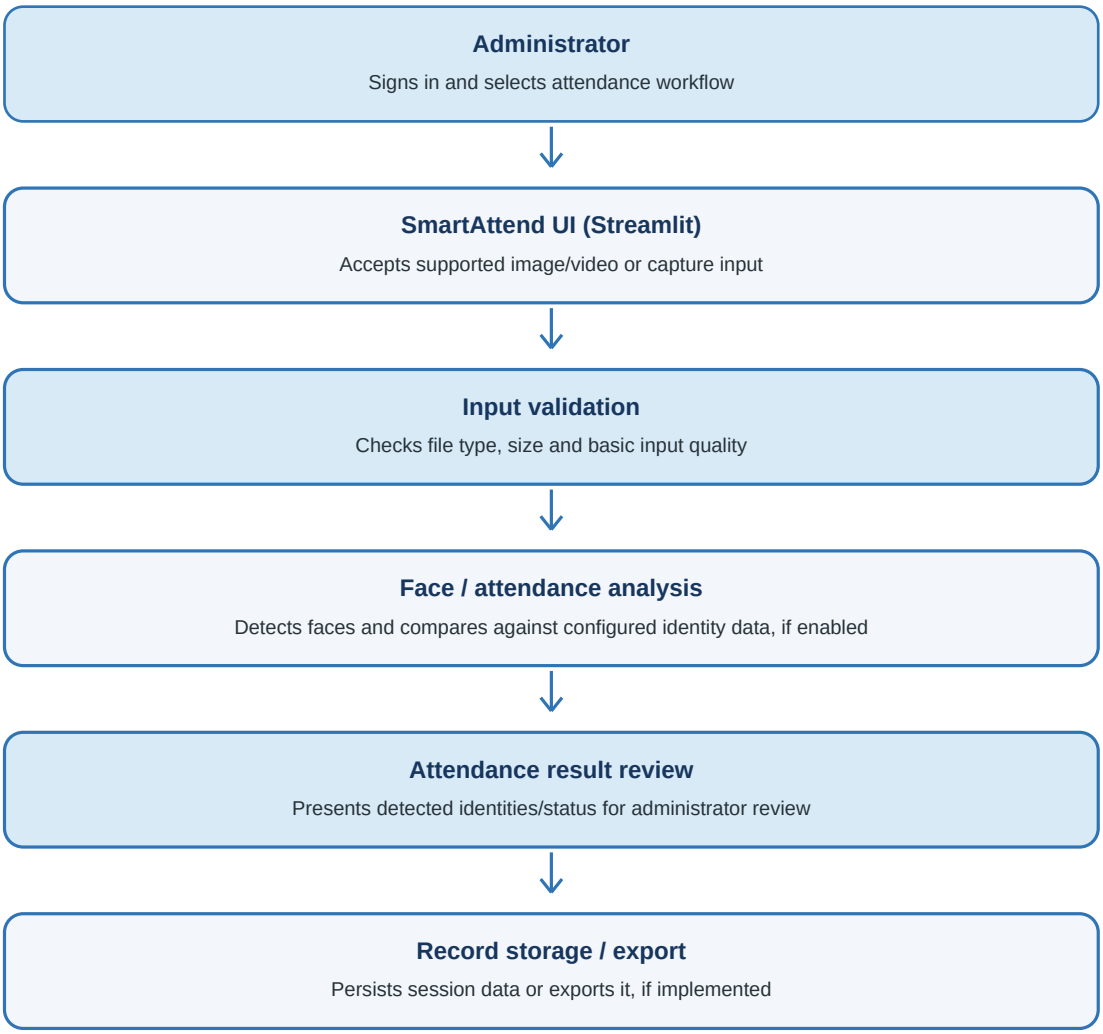
Project documentation • Requirement Analysis

Prepared for an AI-assisted attendance management application

1. Overview

The diagram shows a proposed flow from administrator input to attendance results. Exact storage and recognition components should be adjusted to match the repository implementation.

2. Level 1 data flow



3. Data stores and external entities

Element	Role	Example data
Administrator	External actor	Login credentials; selected class/session
Input media	Input data source	Classroom image/video; file metadata
Recognition component	Processing component	Detected face regions / match confidence
Student or face registry (if configured)	Reference data store	Student identifier and permitted reference image/embedding
Attendance records (if implemented)	Persistent data store	Student/session ID, status, timestamp, review state
Streamlit dashboard	Output interface	Processing status, attendance summary, warnings

4. Data handling and privacy

• Restrict administrative functions to authorized users.
• Avoid storing raw classroom media longer than necessary.
• Protect student identity data and attendance records from unauthorized access.
• Provide a way to review uncertain recognition results instead of treating every match as definitive.
• Follow applicable school/institution consent and privacy policies.

5. Important implementation note

The diagram is a logical proposal. Confirm whether the current SmartAttend code uses local files, a database, Supabase, a webcam, video upload, or an external recognition service before treating those as implemented architecture.